DBMS

Date: ...../...../.....

Functional Dependencies

$\alpha \rightarrow \beta$ (α determines β)

For a relation $r(R)$ this functional dependency holds if and only if for all pair of ~~ty ty~~ tuples t_1 and t_2 of the relation r in the instance is such that :

if, $t_1[\alpha] = t_2[\alpha]$
then, $t_1[\beta] = t_2[\beta]$

NOTE: $\alpha \rightarrow \beta$ is not necessarily equivalent to $\beta \rightarrow \alpha$

Types 5
- → Trivial
- → Non trivial
- → Semi trivial
- → Multivalued (not required)
- → Transitive

a) Trivial FD : # $\alpha \rightarrow \alpha$, $\alpha\beta \rightarrow \alpha$ } Given, no need to check.

b) Non-trival FD : $\alpha \rightarrow \beta$ → we need to check if it holds

c) Semi-trivial FD : $\alpha\beta \rightarrow \beta\gamma$ → we need to check if it holds.

d) Multivalued : $\alpha \twoheadrightarrow \beta$ → α determines the set of values of β.

e) Transitive FD : If, $\alpha \rightarrow \beta$ & $\beta \rightarrow \gamma$ → No need to check
Then, $\alpha \rightarrow \gamma$

Date: ...../...../.....

Closure / F+ :

If F is the given set of FD's given, then F^+, or the closure of F is set containing the FD's of F plus the additional / secondary / inferred FD's from F.

Armstrong's Axioms : used to infer all the functional dependencies on a given ~~relationship~~ relational DB.

1. **Reflexivity :** If, $\beta \subseteq \alpha$,
   then, $\alpha \rightarrow \beta$ (trivial dependency)

2. **Augmentation :** If, $\alpha \rightarrow \beta$,
   then, $\gamma\alpha \rightarrow \gamma\beta$

3. **Transitivity :** If $\alpha \rightarrow \beta$ & $\beta \rightarrow \gamma$
   Then, $\alpha \rightarrow \gamma$

Armstrong's Axioms : Additional Rules :

1. **Union :** If, $\alpha \rightarrow \beta$ & $\alpha \rightarrow \gamma$ holds
   Then, $\alpha \rightarrow \beta\gamma$

2. **Decomposition :** If, $\alpha \rightarrow \beta\gamma$ holds:
   Then, $\alpha \rightarrow \beta$ & $\alpha \rightarrow \gamma$

3. **Pseudotransitivity :** If $\alpha \rightarrow \beta$ & $\gamma\beta \rightarrow \delta$ holds
   Then $\alpha\gamma \rightarrow \delta$

4. **Composition :** If $\alpha \rightarrow \beta$ & $\gamma \rightarrow \delta$ then
   Then $\alpha\gamma \rightarrow \beta\delta$

   ↳ generalization of the union rule.

Date: ...../...../.....

Finding the set of Candidate Keys for a given set of FDs. (Method 3)

Step 1: Start with all the attributes and then eliminate attributes using FDs

Step 2: Now we are left with a ~~key~~ superkey. Check all the proper subsets of this superkey if any ~~or~~ of them are candidate keys. The shortest proper subset to form the superkey will be the candidate key. Else, this superkey is ~~the~~ a candidate key.

Step 3: The attributes in the candidate key form the primary attributes (PA). From the FD's the dependent attributes are termed DA. If $PA \cap DA = \phi$, no other candidate keys are possible.

Step 4: If $PA \cap DA \neq \phi$, and if $PA \cap DA = \alpha$, α is a replaceable attribute. Look for FDs where α is dependend / determined, ie, $\beta \rightarrow \alpha$. Replace α from the original superkey with β and repeat step 2-4 until no more new candidate keys are discovered.

Example:

Given : $R(A, B, C, D, E)$, find all CKs. when FD $\{A \rightarrow B, D \rightarrow E\}$

→ Step 1 : $ABCDE^+ = \{A, B, C, D, E\}$ → Since $A \rightarrow B$ & $D \rightarrow E$,
$ACD^+ = \{A, B, C, D, E\}$ → ACD^+ is a SK

Step 2: Checking proper subsets of ACD for superkey.

$AC^+ = \{A, C, B\}$	×	$A^+ = \{A, B\}$	×
$AD^+ = \{A, D, E, B\}$	×	$C^+ = \{C\}$	×
$CD^+ = \{C, D, E\}$	×	$D^+ = \{D, E\}$	×

∴ ACD is CK.

Date: ...../...../.....

Step 3 : $PA = \{A, C, D\}$
$DA = \{B, E\}$

$PA \cap DA = \phi$. So ACD is the only CK.

Example 2 :

Given : $R(A, B, C, D)$ FD $\{A \to B, B \to C, C \to A\}$

→ Step 1 : $ABCD^+ = \{A, B, C, D\}$ since $A \to B$, ~~$B \to A$~~ $B \to C$
↳ $A \to C$

$AD^+ = \{A, B, C, D\}$ AD^+ is a SK.

Step 2a: $A^+ = \{A, B, C\}$ ×
$D^+ = \{D\}$ × AD is CK.

Step 3a: $PA = \{A, D\}$
$DA = \{A, B, C\}$
$PA \cap DA = \{A\}$ A is replaceable from AD.
Find FD where $\beta \to A$.

$C \to A$ C can replace A.
So, $CD^+ = \{C, D, A, B\}$ CD is a SK.

Step 2b : $C^+ = \{A, B, C\}$ ×
$D^+ = \{D\}$ × CD is the CK.

Step 3b : PA = {A, D, C}

$DA = \{A, B, C\}$
$PA \cap DA =$ {A, C}

↳ Replacing C with B since B -> C

Date: ...../...../.....

$BD^+ = \{B, D, C, A\}$ BD is a SK.

Step 2c: $B^+ = \{B, C, A\}$ ×

$D^+ = \{D\}$ × BD is the CK.

Step 3c: PA = {A, B, C, D}

DA = {A, B, C}

PA ∩ DA = {A, B, C}

can replace B with A since A → B to form the SK AD. But AD was already formed previously so we stop.

Conclusion: CK = {AD, CD, BD}

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R(A,B,C,D)
PA {A, D}
NPA {B, C}
AD -> B (doesnt exist)
AD -> C (doesnt exist)
 therefore 1NF.
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Date: ...../...../.....

Normalization

Process of organizing a DB into well-structured relations based on FDs and keys, to reduce redundancies and anomalies.

- redundancies → same info stored multiple times
- anomalies → unwanted side effect of insert, update, delete caused by redundant or poor table design.

Types of Normalization :

a) First Normal Form (1NF)
b) Second Normal Form (2NF)
c) Third Normal Form (3NF)
d) Boyce-Codd Normal Form (BCNF/3.5NF)
e) Fourth Normal Form (4NF) (not included)

Conditions to satisfy each normal form :

a) 1NF : i) All attributes must be atomic
ii) Each attribute must have a single value

Example :

Name	Address	Phone
Tawfiq, Ahmed	Dhaka, BD, 1215	017, 018
violates (i)	violate (i)	violates (ii)

Date: ...../...../.....

b) 2NF : i) Must be 1NF
ii) Must not contain any partial dependencies.

PSS (CK) → NPA

Example : R(A,B,C), F{AB → C}

a non-prime attribute.

→ CK = AB . Since C is determined by the whole CK and not by any ~~s~~ proper subset of the CK, C is ~~in~~ not partially dependent and the relation is 2NF.

c) 3NF : i) Must be 2NF
ii) Must non contain any transitivity dependency for non prime attributes.

→ NPA → NPA

(i) Example: R(A,B,C,D), F{A→B, B→C, C→D}

→ CK = A . However the FD's B→C, C→D hold which are all non-prime so this relation isnt in 3NF.

(ii) Example : R(A,B,C,D,E,F), FD{AB→CDEF, BD→F}

→ CK = AB . Here, the FD BD→F holds, from which even though B is a PA, ~~E~~D is an NPA so this relation contains NPA→NPA . So its not in 3NF.

d) BCNF: i) Must be 3NF
ii) For each non trivial FD, α→β, α must be a SK.

Example : R(A,B,C), F{A→B, B→C, C→A}

CK = {A, B, C}

all SKs {A → B, B → C, C → A} so this relation is in BCNF.